



S2.1 The ionic model

Structure 2 · Models of bonding and structure · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Ionic bonding results from the electrostatic attraction between oppositely charged ions formed when electrons transfer from a metal (forming a cation) to a non-metal (forming an anion); ions arrange into a giant three-dimensional lattice that maximises attraction between opposite charges and minimises repulsion between like charges. Ionic compounds typically have high melting and boiling points because a large amount of energy (lattice enthalpy) is needed to overcome the strong electrostatic forces throughout the lattice, and lattice enthalpy increases with increasing ionic charge and decreasing ionic radius. Ionic solids do not conduct electricity because ions are fixed in place, but conduct when molten or dissolved in water, where ions become mobile charge carriers.

Worked example: follow the reasoning

Explain why magnesium oxide, MgO , has a much higher melting point than sodium chloride, NaCl .

Step 1: MgO consists of Mg^{2+} and O^{2-} ions; NaCl consists of Na^+ and Cl^- ions.

Step 2: Coulombic attraction between ions is proportional to the product of their charges and inversely related to the distance between their centres.

Step 3: Mg^{2+} and O^{2-} have greater charges (2+ and 2-) than Na^+ and Cl^- (1+ and 1-), and $\text{Mg}^{2+}/\text{O}^{2-}$ are also smaller ions than Na^+/Cl^- , giving a shorter interionic distance.

Answer: MgO has a much stronger lattice (higher lattice enthalpy) due to greater ionic charge and smaller ionic radii, so significantly more energy is required to separate the ions, giving MgO a higher melting point than NaCl .

Quick check · multiple choice

Which property is characteristic of ionic compounds?

- A They conduct electricity in the solid state
- B They are generally soluble in non-polar solvents
- C They conduct electricity when molten or in aqueous solution
- D They have low melting points and are volatile

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Describe the ionic bonding present in calcium fluoride, CaF_2 , and explain why it forms a giant lattice structure rather than a simple molecule. [3] [7]
(b) Predict, with a reason, whether the lattice enthalpy of CaF_2 would be greater or smaller than that of CaCl_2 . [2]
(c) Explain why solid CaF_2 does not conduct electricity, but molten CaF_2 does. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Ca loses 2 electrons to form Ca^{2+} ; each F atom gains 1 electron to form F^- ; the resulting ions are held together by strong electrostatic (ionic) attraction between oppositely charged ions in all directions, so a huge number of ions pack together in a repeating 3-D lattice that maximises attraction and minimises repulsion, rather than forming discrete molecules. [3]

(b) CaF_2 would have the greater lattice enthalpy, because F^- is a smaller ion than Cl^- (fewer electron shells), giving a shorter interionic distance and hence stronger electrostatic attraction for the same ionic charges. [2]

(c) In the solid, ions are held in fixed positions in the lattice and cannot move to carry charge; on melting, the lattice breaks down and the ions become free to move, allowing them to carry charge/conduct electricity. [2]

Quick check answer

C. Ions become mobile charge carriers when the lattice is broken down by melting or dissolving, allowing conduction; solid ionic lattices cannot conduct because ions are fixed in position.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

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